2019-Ph-H1-Q2

Section: Our Dynamic Universe
Topic: Motion, Equations and Graphs

Summary of Question:
A car accelerates uniformly from rest. What happens to the car's **displacement during each second** of motion?

Answer:

- Displacement increases **non-linearly** because velocity increases steadily.
- Each second, the car travels **further** than in the previous second.

✅ **Correct answer: D. It increases by a greater amount each second**

Guidance for Students:

- With constant acceleration, displacement doesn't increase by equal amounts — it accelerates.
- $s = ut + \frac{1}{2}at^2$ shows a quadratic relationship with time.

Revision Tips:

- 📚 Know the SUVAT equations.
- 💡 Displacement grows faster over time under constant acceleration.
- 🔢 The difference in displacement each second gets larger.